Apparatus for Measurements

Exam code: 0620 & 0971

Download PDF

Written by: Stewart Hird
Reviewed by: Lucy Kirkham
Updated on 30 September 2025

Time, temperature, mass & volume

Time

- Time can be measured using a **stopwatch** or stopclock which are usually accurate to one or two decimal places
- The units of time normally used are **seconds** or **minutes**
  - Other units may be used for extremely slow reactions (e.g. rusting)
- **Remember:** 1 minute = 60 seconds

Examiner Tips and Tricks

Careful: Units of time often cause issues in results tables.

If the display on a stopwatch showed 1:30.

- The incorrect time to record would be 1.30 minutes.
- The correct time would be 1.5 minutes.

To avoid any confusion, if the time intervals are less than a minute, it is best / easier to change the recorded units to seconds.

- So, the same stopwatch display would be recorded as 90 seconds.

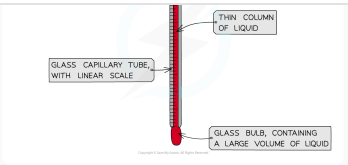
Temperature

- Temperature is measured with a **thermometer** or digital **temperature probe**

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

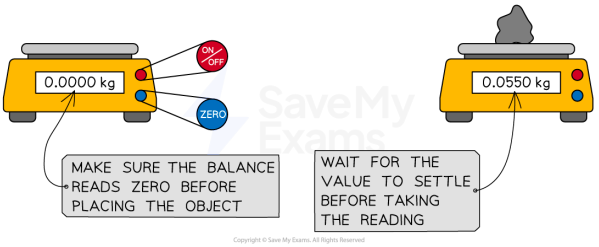


- Laboratory thermometers usually have a precision of a half or one degree
  - Digital temperature probes are available which are more **precise** than traditional thermometers and can often read to 0.1 °C
- Traditional thermometers rely upon the uniform expansion and contraction of a liquid substance with temperature
  - Digital temperature probes can be just as, if not, more **accurate** than traditional thermometers
- The units of temperature are **degrees Celsius (°C)**

Help me understand

Mass

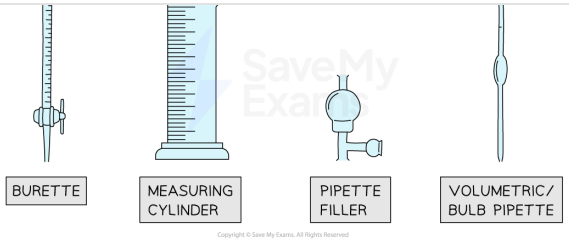
- Mass is measured using a **digital balance** which normally gives readings to two decimal places
  - Balances should be tared (set to zero) before use
  - Balances should also be allowed time to settle on a final measurement / reading before it is recorded



- The standard unit of mass in kilograms (kg)
  - However, in chemistry **grams (g)** are most often used
- **Remember:** 1 kilogram = 1000 grams

Volumes of liquid

- The volume of a liquid can be determined using different pieces of apparatus
  - The choice of apparatus depends on the level of accuracy needed
- Three common pieces of apparatus for measuring the volume of a liquid are:
  - **Burettes**
  - **Volumetric pipettes**
  - **Measuring cylinders**



- **Burettes** are the most accurate way of measuring a **variable** volume of liquid between 0 cm^3 and 50 cm^3
  - They are most commonly used in titrations
  - **Careful:** Read the burette scale from top to bottom as 0.00 cm^3 is at the top of the column
- **Volumetric pipettes** are the most accurate way of measuring a **fixed** volume of liquid,
  - They have a scratch mark on the neck which is matched to the bottom of the meniscus to make the measurement
  - A pipette filler is used to draw the liquid into the volumetric pipette
  - The most common volumes for volumetric pipettes are 10 cm^3 and 25 cm^3
- **Measuring cylinders** are used when approximate volumes are required (accuracy is not an important factor)
  - These are graduated (have a scale so can be used to measure)
  - Measuring cylinders typically range from 10 cm^3 to 1 litre (1 dm^3)
- Whichever apparatus you use, you may see markings in millilitres, ml, which are the same as a cm^3

Volumes of gas

- For some experiments, the volume of a gas produced needs to be measured
- This is typically done by using one of the following methods:
  - Using a **gas syringe**
  - By **downward displacement of water**
- A **gas syringe** is more precise and accurate than downward displacement of water

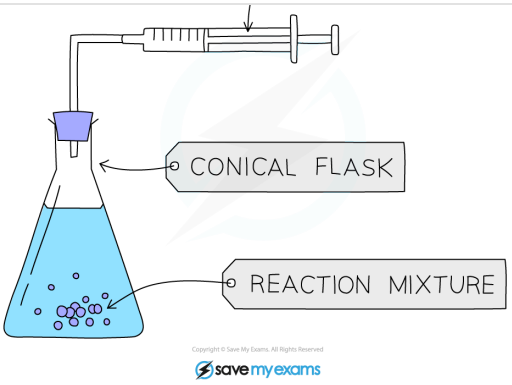


Diagram of the set-up for an experiment involving a gas syringe

- **Downward displacement of water** is where a **measuring cylinder** is inverted in water to collect the gas produced
  - This method does not work if the gas is soluble in water

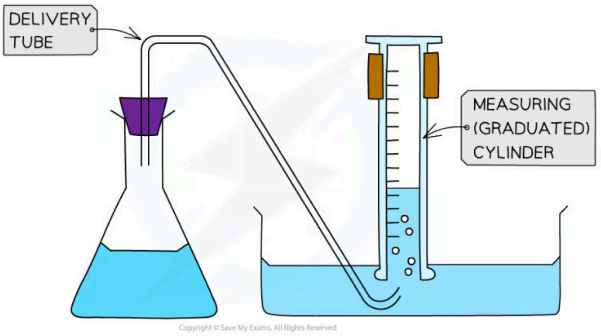


Diagram of the set-up for an experiment collecting gas by downward displacement of water

- If the gas happens to be heavier than air and is coloured, the cylinder does not need to be inverted

Advantages & disadvantages of methods & apparatus

- In the lab, we often have choices of different apparatus to do the same job
- Evaluating which piece of apparatus is the best one to use is part of good experimental planning and design
- This means appreciating some of the advantages and disadvantages of laboratory apparatus

Advantages and disadvantages of lab apparatus

Temperature probe	• More precise readings • Easy to make multiple repeat readings • Can be automated to run over long periods of time	• Can be corroded by some reagents ○ More expensive (to replace)
Volumetric pipette	• Accurate measurement of a fixed volume	• Harder to use than a normal pipette • Only measures one fixed volume
Gas syringe	• Easy to set up • Keeps the gas dry	• The syringe can stick • Collects limited volumes • Expensive and delicate / fragile
Microscale experiments	• Less wasteful • Saves energy • Safer	• Hard to see what's happening • Lose a lot of material separating / purifying the products

Five pieces of apparatus that can be used to measure the volume of a liquid. They all have their pros and cons

Planning your method

- Good experimental design includes the answers to questions like
  - Have I chosen a suitable apparatus for what I need to measure?
  - Is it going to give me results in an appropriate time frame?
  - Is it going to give me enough results to process, analyse and make conclusions?
  - Does it allow for repetitions to check how **reliable** my results are?
  - Does my plan give a suitable **range** of results?

- You may be asked about experimental methods in exam questions and your experience and knowledge of practical techniques in chemistry should help you to spot mistakes and suggest improvements

Examiner Tips and Tricks

Make sure you know the names of common laboratory apparatus.

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself

Flashcards

Was this revision note helpful? Yes No

Previous

Next

Course

Study tools

All topics

Resources

Learning Hub

Scholarship

Join

Past Papers

Solution Banks

Dashboard

Overview

Brand ambassador

Profile

Subscription

Settings

Company

About us

Exam Specificity

Content Quality

Promotions

Jobs

Terms

Privacy

Cookie Policy

Help and Support

Subjects

Biology

Chemistry

Physics

Maths

Geography

English Literature

Psychology

All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.